■ 4.2.6 APL

You can use *Mathematica* very much like APL. You can represent APL arrays as nested *Mathematica* lists. Most mathematical operations in *Mathematica*, as in APL, can operate on arrays of elements as well as on single elements.

APL operation	*related Mathematica function*
subscripting a[*i*]	a[[*i*]] (*i* can be a list)
index (ι)	`Range`
shape (ρ)	`Dimensions`
ravel (,)	`Flatten`
reshape (ρ)	`Partition`
catenate (,)	`Join`
membership (ϵ)	`MemberQ`
take ($\uparrow$) and drop ($\downarrow$)	`Take` and `Drop`
reverse and rotate	`Reverse` and `RotateLeft`
transpose (monadic and dyadic)	`Transpose`
grade up	roughly like `Sort`
encode and decode	roughly like `^^` and `BaseForm`
compress (/)	`Select`
outer product ($\circ . f$)	`Outer[`*f*`, ...]`
inner product ($f.g$)	`Inner[`*g*`, ..., `*f*`]`
reduce (f/)	`Apply[`*f*`, ...]`

Some correspondences between APL and *Mathematica*.

The symbolic nature of *Mathematica* allows you to give "operations" or "functions" as arguments to other functions. In this way, you can effectively create composite operators. Thus, for example, the APL composite operator `+.*` becomes `Inner[Times, `*a*`, `*b*`, Plus]` in *Mathematica*.

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